

# LongCat-DeepResearch

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## ABSTRACT

We introduce LongCat-DeepResearch, which combines a LongCat model enhanced for deep research with a research harness organized around an executable ResearchSpec. Open-ended research requires investigation to adapt as analysis reveals new information needs. The harness shifts early iteration from full reports to a compact specification of research requirements, reducing reliance on repeated full-report rewriting. Multiple planners independently explore external sources, then consolidate and refine questions, evidence requirements, and section responsibilities. Researchers investigate and draft their assigned sections in separate contexts, gathering further evidence as their analyses develop; a Global Editor then coordinates local revisions to the assembled report. These stage interfaces also support research-task and trajectory construction for LongCat’s general-purpose model training. LongCat-DeepResearch achieves 55.25 on DeepResearchBench, 51.35 on DeepResearchBench II, and 79.83 on ResearchRubrics. It scores 76.04 on the in-house benchmark, ranking second among the four compared systems. Development-set analyses show that aggregating planning perspectives improves the observed report scores, while further refinement has mixed effects. Enhanced Editor rounds improve average automatic readability preference across two benchmarks, with different trends within each benchmark.

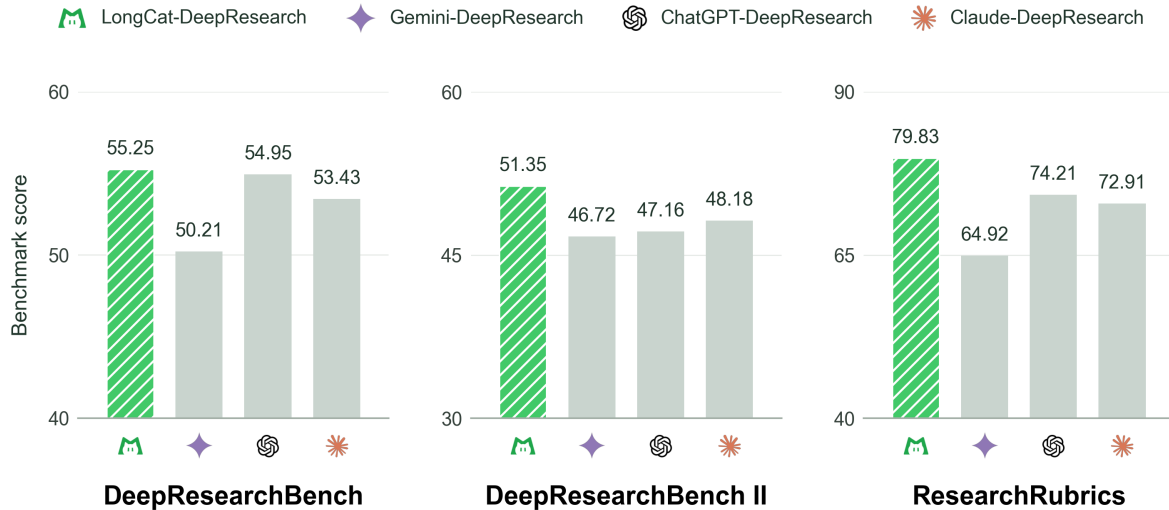


Figure 1: Public-benchmark scores.

## 1 Introduction

Open-ended deep research requires language-model agents to investigate underspecified questions and gather evidence from diverse external sources [Shao et al., 2024, Chen et al., 2024, Li et al., 2025]. These agents synthesize their findings into comprehensive long-form reports [Tongyi DeepResearch Team, 2025, Li et al., 2026c, Zhu et al., 2026b, Li et al., 2026b, Zhang et al., 2026b]. Unlike conventional information seeking, the evidence requirements of such reports cannot be fully specified before the investigation begins. As an analysis develops, emerging explanations expose missing evidence, unresolved claims require further verification, and new research questions arise. A research system must therefore establish a useful initial agenda while allowing further evidence gathering as its sections develop. This

raises a practical question: *How can deep-research agents coordinate a shared research agenda while preserving the detailed investigation needed to write each section?*

Existing systems instantiate this research–writing feedback in different ways. Prior approaches interleave reasoning, retrieval, and drafting [Li et al., 2025], or use an evolving report to guide subsequent retrieval and revision [Han et al., 2025]. Other approaches organize investigation through multiple perspectives and evidence-grounded outlines before detailed composition [Shao et al., 2024, Li et al., 2026c]. When this interaction is carried by a growing research history or repeated full-report updates, three difficulties arise. First, evidence, intermediate reasoning, and draft text compete for finite context; compression may remove information whose relevance has not yet become apparent. Second, early findings can anchor the developing narrative and steer subsequent searches, narrowing the perspectives explored. Third, updating the full report repeatedly regenerates text beyond the passages affected by new evidence. The challenge is to retain feedback between research and writing without concentrating the entire investigation in an expanding draft.

In this work, we introduce LongCat-DeepResearch, a system combining an improved LongCat model with a research harness centered on an executable *ResearchSpec*. The key idea is to shift early iteration from the full report to a compact specification of its research requirements. Multiple planners independently search and read external sources to develop candidate specifications. Their proposals are consolidated and refined to identify missing questions, evidence requirements, and section responsibilities. This process encodes the emerging global understanding of the task into *ResearchSpec*, which serves as a compact proxy for what the report needs to establish. The downstream synthesis pipeline retains complete section drafts through assembly and editing. Each researcher receives the complete *ResearchSpec* and one assignment, continues gathering evidence in an independent context, and writes a complete section from its findings. The resulting section artifacts are assembled into a draft, after which a Global Editor identifies cross-section issues and Local Editors perform targeted revisions. This design combines compact global coordination with detailed local research. *ResearchSpec* is refined before section dispatch; subsequent investigation develops within the assigned sections, and editing coordinates their text without reopening the global research agenda.

We evaluate LongCat-DeepResearch on three established benchmarks and an in-house benchmark. It achieves 55.25 on DeepResearchBench, 51.35 on DeepResearchBench II, and 79.83 on ResearchRubrics, outperforming the strongest of the three compared deep-research products by 0.30, 3.17, and 5.62 points, respectively, in the recorded system-level comparison (Section 4.3). On our in-house benchmark, LongCat-DeepResearch ranks second with an overall score of 76.04, exceeding Claude-DeepResearch at 61.42 and Gemini-DeepResearch at 42.49, and trailing ChatGPT-DeepResearch’s 76.59 by 0.55 points. The same stage interfaces also support the construction of research questions, task-specific rubrics, and trajectories used in the mid-training and post-training of LongCat’s general-purpose models.

## 2 Related Work

**Context management and evidence preservation.** Long-context access does not ensure reliable use of evidence throughout the input [Liu et al., 2023]. MemGPT separates working context from external storage [Packer et al., 2023], while AgentFold learns fine-grained context folding and RE-TRAC carries structured summaries across research attempts [Ye et al., 2025, Zhu et al., 2026c]. FoldAct studies the training instability introduced when generated summaries change future observations [Shao et al., 2025]. AdaCoM further shows that the preferred degree of compression varies with the underlying agent [Yi et al., 2026]. These works make compression an explicit, useful operation rather than an incidental implementation detail. In deep research, SearchSwarm returns compact, citation-grounded subagent reports to an orchestrator [Lan et al., 2026], and Argus synthesizes an answer from a compact evidence graph [Zhang et al., 2026c]. Deep-Reporter maintains a recurrent global summary and a verbatim local tail during sequential section generation [Ye et al., 2026]. Our report-synthesis path retains complete section artifacts through assembly and global-to-local editing, while external memory and compression address the separate management of research history.

**Research planning and structured state.** WebGPT and ReAct established tool-grounded interaction [Nakano et al., 2021, Yao et al., 2022]. STORM develops perspectives before article writing, and MindSearch decomposes search through an executable graph [Shao et al., 2024, Chen et al., 2024]. OmniThink uses an information tree and a conceptual pool to connect knowledge expansion with reflection [Xi et al., 2025]. More recent work makes the research structure mutable: WebWeaver interleaves evidence acquisition with outline optimization, AgentCPM-Report alternates drafting and deepening, and ScaffoldAgent selects outline changes using downstream utility [Li et al., 2026c,b, Yang et al., 2026]. Enterprise Deep Research combines coverage objectives, dependency-guided information sharing, and evidence-sufficiency conditions [Choubey et al., 2026]; SearchOS externalizes coverage, evidence, pending tasks, and failed searches [Zhang et al., 2026b]. DecomposeR makes a typed research DAG trainable with structure-aware rewards [Hussain et al., 2026].

The closest structured-state comparisons also include RhinoInsight, whose verifiable checklists constrain research actions and whose evidence audit links sources to drafted content [Lei et al., 2025], and DualGraph, which maintains separate, co-evolving knowledge and outline graphs [Shi et al., 2026]. These systems establish structured planning and evidence binding as existing design choices. In our pipeline, ResearchSpec defines section-level execution responsibilities that are fixed after planning; researchers retain their findings in section artifacts passed to editorial reconciliation. Its role is an execution interface, rather than a persistent knowledge graph or an input-level specification for personalized query refinement [Yoon and Lee, 2026].

**Long-form synthesis and revision.** PaperQA2 develops cited scientific-topic summaries and literature-contradiction detection [Skarlinski et al., 2024]; OpenScholar combines literature retrieval, citation-backed generation, and iterative self-feedback [Asai et al., 2026]. These systems establish scientific synthesis as a research objective beyond isolated fact retrieval. In open-ended report writing, WebThinker interleaves reasoning, search, and drafting [Li et al., 2025], while TTD-DR uses an evolving draft to guide retrieval and iterative report revision [Han et al., 2025]. FS-Researcher separates persistent evidence collection from multi-session report writing and retains original source files [Zhu et al., 2026b]. CogGen coordinates a global plan–write–review loop with section-level work and permits global restructuring [Tian et al., 2026]; Ptah maintains inspectable research artifacts and visual memory for multimodal composition [Zhang et al., 2026a]. Our design assembles the independently written sections before deciding how to reconcile them: the Global Editor specifies ownership, while Local Editors revise assigned units. This scope differs from regenerating the entire document in one invocation, but does not guarantee that edits retain every useful detail. Mr. Dre provides direct evidence that report revision can satisfy new feedback while damaging earlier coverage or citation quality [Chen et al., 2026]. DeepTRACE audits statement-level support and attribution, distinguishing listed sources from supported claims [Venkit et al., 2025]. Self-Refine, DuMate, and AREX offer complementary feedback and verification loops [Madaan et al., 2023, Yan et al., 2026, Lu et al., 2026].

**Synthetic research tasks and trajectories.** OpenAI’s Deep Research System Card describes reinforcement learning on browsing tasks that include open-ended tasks graded with rubrics [OpenAI, 2025]. DR Tulu develops an open long-form research training approach in which rubrics evolve with the policy and newly acquired evidence [Shao et al., 2026]. These are distinct precedents: rubric-based supervision predates the evolving-rubric formulation. Tongyi DeepResearch and Step-DeepResearch describe synthetic tasks and agentic training across mid-training and post-training [Tongyi DeepResearch Team, 2025, Hu et al., 2025]. S1-DeepResearch emphasizes planning, evidence integration, and report generation beyond search-centric QA [Dong et al., 2026], while Marco DeepResearch emphasizes verification in task and trajectory construction [Zhu et al., 2026a]. For open-ended reports, requirements must be supported and aligned with the question. ResearchRubrics contributes expert-written prompts and criteria, including implicit requirements and negative criteria [Sharma et al., 2025]; DeepResearchBench II derives atomic criteria from expert articles [Li et al., 2026a]. These evaluation resources are distinct from synthetic training data. DeepRubric constructs evidence trees and jointly synthesizes queries and rubrics, whereas Quest uses rubric trees to construct both objective and open-ended research tasks [Zhu et al., 2026d, Xie et al., 2026]. Our data discussion builds on these principles and describes document provenance, question–rubric alignment, and trajectory interfaces for research-oriented training data.

### 3 Method

The problem is to coordinate detailed evidence gathering and report construction across separate research contexts. LongCat-DeepResearch addresses this by assigning section research to independent contexts and separating global editorial decisions from local text generation. Its intermediate artifacts also provide units for data construction and stage-wise evaluation.

#### 3.1 Research Harness

ResearchSpec assigns research responsibilities before detailed evidence accumulates. Independent Researchers develop the assigned sections; assembly and editing then coordinate those sections into a report (Fig. 2). Retaining these section artifacts does not imply preserving every retrieved page or every detail of the underlying interactions.

**Explore before committing to a plan.** Independent Planning Writers (planners) briefly search and read sources before proposing the report’s structure, grounding the research scope in available evidence beyond the model’s prior knowledge. Their separate explorations can reveal complementary perspectives, as in perspective-guided article planning [Shao et al., 2024]. The Planning Judge consolidates the candidate specifications, the Critic searches for missing questions and cases, and the Reviser incorporates the feedback. Because section assignments direct subsequent research, this

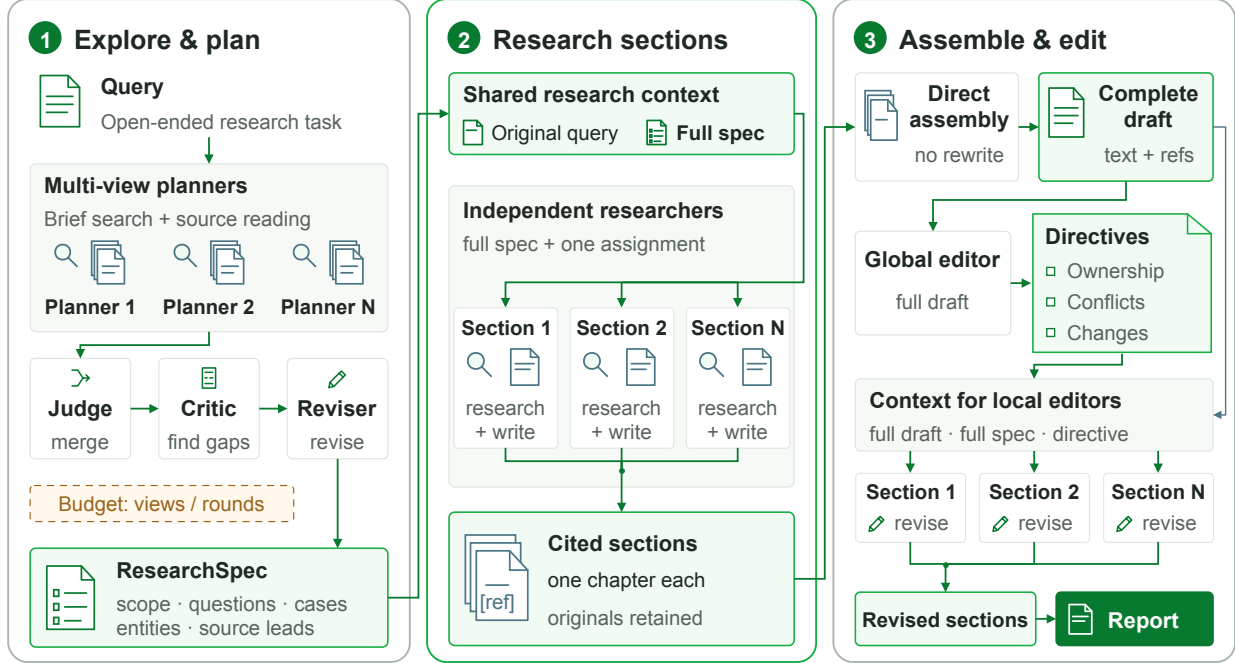


Figure 2: The LongCat-DeepResearch harness. Planning produces a shared ResearchSpec; independent Researchers produce complete section drafts; global editorial directives guide local revisions after assembly. ResearchSpec is fixed during section research, and both editor types receive the full assembled draft.

refinement seeks to identify omissions before a direction is left uninvestigated. Additional candidates and refinement rounds provide ways to allocate more computation to planning; their benefit is not guaranteed.

ResearchSpec records each section’s scope, research questions, required entities or cases, and provisional source leads. This compact representation of the intended report also specifies how research work is divided across contexts. Its coverage and responsibilities can be inspected and revised before generating a full draft. Each dispatchable subsection has a unique hierarchical ID, such as S1.2; we refer to these execution units as sections. Before dispatch, validators check ID uniqueness, parent relationships, consecutive ordering, and required fields. Invalid plans undergo repair or revert to a validated planning candidate; unresolved validation errors stop dispatch. Findings and source leads must still be established through subsequent research and verification. The specification is revised within the planning stage and then held fixed during section research; a later research round that reopens the plan is an extension beyond the current pipeline.

**Research and write in independent section contexts.** Each Researcher receives the original query, the complete ResearchSpec, and one section assignment. The global specification establishes how its local work contributes to the report, while separate contexts accommodate the branches’ detailed tool interactions. Researchers receive the shared specification, without depending on previously generated sections, and can extend their investigation while addressing the assigned requirements. They execute concurrently and write their own citation-bearing sections from the evidence they have acquired. The same agent therefore carries its local evidence into writing without first compressing it into a summary for a separate writer. The complete sections retain the developed arguments and details, reducing the reliance of subsequent composition on a repeatedly compressed shared history.

**Assemble first, then coordinate and edit.** The harness mechanically assembles sections in ResearchSpec order, retaining their text and citations even when content overlaps. A Global Editor reads the complete draft, assigns ownership of repeated material, identifies conflicts, and specifies changes. Local Editors apply these directives to their assigned sections using the complete draft and specification as context, drawing on facts and citations already present in the draft. Their output scope is local, so one editorial call need not regenerate the entire document. This separates the global judgment needed for coordination from the generation of revised text, while keeping the original sections and assembled draft available for comparison. Both editor types read the full draft and remain subject to input-context

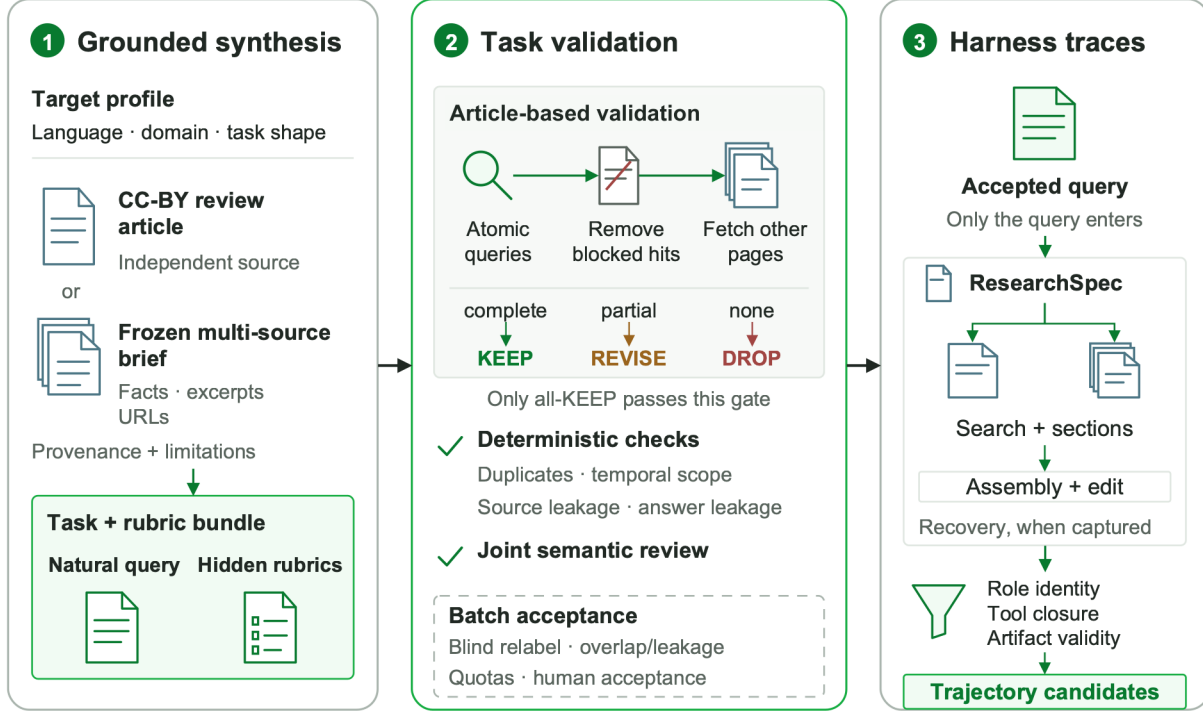


Figure 3: Evidence-grounded construction of research tasks and trajectories. Questions and rubrics are constructed from independent source material; only the query is supplied to the answering agent.

limits. Local output scopes reduce the text regenerated by each call; repeated full-draft inputs can still incur substantial cost. Research and editing can omit useful details.

**Decoupled interfaces for data construction and evaluation.** Planning maps the query and explored sources to a ResearchSpec; research maps section assignments to grounded sections; editing maps a draft and directives to revised sections. These mappings provide concrete units for targeted data synthesis and rubric-based evaluation. ResearchSpecs can be checked for coverage before detailed investigation, sections against their assigned requirements, and revisions against editorial directives and original text. The decomposition supports separate improvement and computation allocation at each stage, and provides the foundation for the research-data construction described next.

### 3.2 Research Data Construction

Data construction follows the harness’s stage interfaces: we first build research tasks with evidence-backed requirements, then collect trajectories that show how those tasks are planned, investigated, and edited. Independent source material grounds the questions and rubrics used in this process.

**Ground questions and rubrics in source material.** A target profile specifies language, topic, breadth, and the intended report. The article-based route starts from an independently licensed review article; a complementary route uses frozen multi-source briefs containing facts, excerpts, URLs, and source limitations. Both routes organize evidence for questions that require explanation or comparison across sources.

Each question is paired with task-specific rubrics. Following evidence-grounded synthesis principles [Zhu et al., 2026d, Xie et al., 2026], factual criteria must have supporting evidence, while analytical criteria specify warranted comparisons or inferences. The rubric can include implicit requirements and negative conditions, as distinguished in ResearchRubrics [Sharma et al., 2025]. Construction evidence and rubrics remain separate from the query supplied to the answering agent.

**Check evidence, searchability, and task quality.** The article-based route supports a bounded searchability check for atomic information-recall criteria. It searches for alternative sources, removes hits from the excluded construction

article, fetches the remaining pages, and checks support for the complete factual requirement. Complete, partial, or missing support produces KEEP, REVISE, or DROP; the gate passes only when all retained criteria receive KEEP. Deterministic checks address duplicate rubrics, answer leakage, excluded-source leakage, and temporal scope. Joint semantic review checks whether the question, evidence, and rubric describe a coherent research task. The bounded search tests evidence availability within its budget, without certifying exhaustive Web answerability.

**Collect and filter stage-specific trajectories.** Accepted queries drive teacher executions of the harness, recording candidate and final ResearchSpecs, tool requests and observations, citation-bearing sections, assembled drafts, and editorial decisions. Parallel branches retain their actual inputs and dependencies. Filtering checks role identity, tool-call/response closure, and intermediate-artifact validity, while preserving distinctions among complete, partial, and failed attempts. Each record identifies its teacher and harness. Trajectory validity is one part of dataset selection: dataset-level overlap checks, distribution-aware selection, and human review where specified are additional checks before training-data acceptance.

**Model training.** Research-related data are used alongside other data in the mid-training and post-training of LongCat’s general-purpose models to improve deep-research capabilities. LongCat-DeepResearch combines this model with the research harness described in [Section 3.1](#). The system-level scores do not isolate the contribution of this pipeline, one data source, or one training stage.

## 4 Evaluation

We organize evaluation around four benchmarks: DeepResearchBench, DeepResearchBench II, ResearchRubrics, and an in-house benchmark. Following each benchmark’s official evaluation setup, DeepResearchBench RACE and DeepResearchBench II are judged by GPT-5.5 (medium), while ResearchRubrics is judged by Gemini 2.5 Pro. These models serve as evaluators of the generated reports. Evaluator-specific reference results are available from the [official benchmark leaderboard](#). The official snapshots and the four systems evaluated here are presented together in [Appendix A](#). The first three provide the recorded comparison with Gemini-DeepResearch, ChatGPT-DeepResearch, and Claude-DeepResearch. The in-house benchmark is reported in [Section 4.5](#). Under the evaluation protocol, the generation input is the original task prompt; benchmark rubrics and reference reports are reserved for scoring. Public-Web retrieval remains available during generation.

### 4.1 Benchmarks

**DeepResearchBench.** DeepResearchBench contains 100 PhD-level research tasks spanning 22 domains, balanced between 50 Chinese and 50 English tasks [Du et al., 2025]. Its RACE protocol assesses each report along four fixed top-level dimensions: *Comprehensiveness*, *Insight/Depth*, *Instruction-Following*, and *Readability*. RACE generates task-specific criteria and weights within these dimensions, scores both a target and a strong reference, and computes the final relative score as  $100 \times S_{\text{tgt}} / (S_{\text{tgt}} + S_{\text{ref}})$  on the percentage scale, where  $S_{\text{tgt}}$  and  $S_{\text{ref}}$  denote the corresponding target and reference scores. Relative scores are computed within each task before macro-averaging across tasks. We report dimension scores where they are available; the current LongCat-DeepResearch row includes all 100 tasks from the September 14 evaluation. For this run, we use the benchmark’s GPT-5.5 (medium) evaluator branch with its official prompt, dynamic criteria, reference reports, weights, and aggregation code unchanged. The reference reports are the April 2025 snapshot generated by Gemini-DeepResearch. DeepResearchBench also provides FACT, which extracts statement–URL pairs, removes duplicate claims associated with the same URL, and checks whether the retrieved page supports each statement. The official current pipeline uses GPT-5.4 Mini for extraction and support judgment and Jina Reader to obtain page text. These quantities should be accompanied by output length because effective-citation counts are length-sensitive. We do not report FACT results. FACT measures statement–page support; it does not by itself establish source authority or the truth of every claim.

**DeepResearchBench II.** DeepResearchBench II contains 132 tasks across the same 22-domain taxonomy, with 66 tasks in each language [Li et al., 2026a]. The benchmark is grounded in expert-written investigative articles. The benchmark paper reports 9,430 atomic rubrics constructed through automatic extraction, self-evaluation, manual revision, and more than 400 hours of expert review. Its rubrics cover *Information Recall*, *Analysis*, and *Presentation*. A rubric passes only when the report contains the required fact or inference; numerical requirements must be matched explicitly. We use the official evaluator and report the pass rate for each dimension together with the official overall score (the mean of task-level pass fractions across all rubrics). Task rubrics and reference text are excluded from the actor prompt by this protocol. If a supporting sentence cites a blocked source article, the corresponding rubric receives the official score of  $-1$ . The product comparison uses GPT-5.5 (medium) as the judge.

**ResearchRubrics.** ResearchRubrics pairs 101 realistic prompts with 2,593 expert-written criteria covering explicit and implicit requirements, synthesis, citation quality, instruction following, and communication quality [Sharma et al., 2025]. We compute a weighted score per task before macro-averaging across tasks. The LongCat result covers all 101 tasks. Dimension scores use the sum of score times weight in the numerator and only positive weights in the denominator; negative-weight criteria contribute penalties to the numerator. Dimension scores are not averaged to obtain the overall result. We use Gemini 2.5 Pro as the judge, following the benchmark’s evaluation setup, and retain its original expert-written criteria and weighted scoring convention. This benchmark complements DeepResearchBench II by testing adherence to prompt-specific criteria that include both requested content and requirements inferred from task context.

**In-house benchmark.** The in-house comparison evaluates LongCat-DeepResearch, Gemini-DeepResearch, ChatGPT-DeepResearch, and Claude-DeepResearch on the in-house benchmark using an automatic evaluator. The resulting dimension scores and weighted overall results are reported in [Section 4.5](#).

## 4.2 Comparison Systems

The comparison includes LongCat-DeepResearch, Gemini-DeepResearch, ChatGPT-DeepResearch, and Claude-DeepResearch. The latter three are accessed through their respective official clients using each provider’s own Deep Research system, including its native tools and budgets. The selected client models are Gemini 3.7 Flash, GPT-5.6 Sol (xhigh), and Claude Opus 5 (xhigh), respectively. The product names in the result tables refer to these complete Deep Research systems with the specified client model selections. These system-level comparisons do not isolate a model or harness component. The configuration study in [Section 5.4](#) compares the previous LongCat release with ReAct/direct writing and the current harness, and compares the previous release with the current LongCat model under the current harness. All three configurations use the full benchmarks.

## 4.3 Evaluation Protocol

The benchmark task sets contain 100 DeepResearchBench tasks, 132 DeepResearchBench II tasks, and 101 ResearchRubrics tasks. DeepResearchBench and DeepResearchBench II use GPT-5.5 (medium), and ResearchRubrics uses Gemini 2.5 Pro. Each benchmark’s overall score is computed per task and then averaged over its scored tasks; ResearchRubrics retains signed criterion weights and a positive-weight denominator. Scored coverage varies across systems. We compare the systems as deployed, including their native research tools and budgets. Detailed coverage and execution records are provided in [Appendix B](#).

## 4.4 Main Results

[Table 1](#) reports the current LongCat-DeepResearch comparison. LongCat-DeepResearch obtains 55.25 on DeepResearchBench, 51.35 on DeepResearchBench II, and 79.83 on ResearchRubrics. Relative to the strongest of the three listed web products, these are differences of +0.30, +3.17, and +5.62 points, respectively. LongCat-DeepResearch has the highest overall scores on these three public benchmarks among the compared systems. These point estimates use the scored cohorts listed below and do not establish statistical significance. The dimension scores show where the observed differences occur and where weaknesses remain.

**DeepResearchBench.** LongCat-DeepResearch leads the overall comparison with 55.25, exceeding ChatGPT-DeepResearch by 0.30 points. It also has the highest scores for comprehensiveness (56.21), insight (56.48), and instruction following (55.18). Readability remains weaker: its 49.96 is below ChatGPT-DeepResearch’s 51.51 and Claude-DeepResearch’s 51.35.

**DeepResearchBench II.** LongCat-DeepResearch scores 51.35, exceeding Claude-DeepResearch by 3.17 points. It leads in information recall (46.42) and analysis (60.69), while its presentation score of 83.13 trails ChatGPT-DeepResearch’s 86.98. The strongest dimension-level advantage is in analysis: 60.69 versus 52.26 for the next-highest system.

**ResearchRubrics.** LongCat-DeepResearch scores 79.83, exceeding ChatGPT-DeepResearch by 5.62 points. It leads in explicit requirements (86.97), implicit requirements (79.52), and synthesis (78.92). Instruction following, citation quality, and communication each remain below at least one compared system. These category scores describe complementary aspects of report quality; their simple average does not reproduce the full-rubric total.

The case study in [Section 5.5](#) illustrates how ResearchSpec, section research, and editing interact in one recorded report.

| Metric                      | Gemini-DeepResearch | ChatGPT-DeepResearch | Claude-DeepResearch | LongCat-DeepResearch |
|-----------------------------|---------------------|----------------------|---------------------|----------------------|
| <b>DeepResearchBench</b>    |                     |                      |                     |                      |
| Comprehensiveness           | 49.47               | 55.41                | 53.71               | <b>56.21</b>         |
| Insight / Depth             | 50.73               | 55.82                | 53.87               | <b>56.48</b>         |
| Instruction following       | 50.88               | 54.99                | 53.59               | <b>55.18</b>         |
| Readability                 | 49.28               | <b>51.51</b>         | 51.35               | 49.96                |
| <b>Overall</b>              | 50.21               | 54.95                | 53.43               | <b>55.25</b>         |
| <b>DeepResearchBench II</b> |                     |                      |                     |                      |
| Information recall          | 41.75               | 41.97                | 44.37               | <b>46.42</b>         |
| Analysis                    | 51.28               | 52.26                | 51.15               | <b>60.69</b>         |
| Presentation                | 84.81               | <b>86.98</b>         | 81.02               | 83.13                |
| <b>Overall</b>              | 46.72               | 47.16                | 48.18               | <b>51.35</b>         |
| <b>ResearchRubrics</b>      |                     |                      |                     |                      |
| Explicit requirements       | 76.36               | 83.57                | 79.64               | <b>86.97</b>         |
| Implicit requirements       | 60.25               | 70.36                | 70.57               | <b>79.52</b>         |
| Instruction following       | 62.39               | <b>74.47</b>         | 74.22               | 66.27                |
| Citation quality            | 32.71               | 64.43                | <b>67.27</b>        | 65.00                |
| Synthesis                   | 60.41               | 69.14                | 70.29               | <b>78.92</b>         |
| Communication               | 59.64               | <b>64.24</b>         | 53.33               | 60.00                |
| <b>Overall</b>              | 64.92               | 74.21                | 72.91               | <b>79.83</b>         |
| <b>In-house</b>             |                     |                      |                     |                      |
| Content                     | 71.66               | 87.10                | 80.83               | <b>87.11</b>         |
| Analysis                    | 18.61               | 66.58                | 44.07               | <b>70.74</b>         |
| Presentation                | 21.35               | <b>74.64</b>         | 53.20               | 52.99                |
| <b>Overall</b>              | 42.49               | <b>76.59</b>         | 61.42               | 76.04                |
| Descriptive average         | 51.09               | 63.23                | 58.99               | <b>65.62</b>         |

Table 1: Public and in-house benchmark results with benchmark-specific dimensions; higher is better. In-house scores use automatic evaluation (Section 4.5); Overall is the weighted total. The descriptive average is the unweighted mean of the four Overall scores, which use different scoring scales and protocols. The Gemini, ChatGPT, and Claude results are obtained through their official clients using each provider’s own Deep Research system, with Gemini 3.7 Flash, GPT-5.6 Sol (xhigh), and Claude Opus 5 (xhigh) selected, respectively. See Appendix B for evaluation details.

#### 4.5 In-house Benchmark

We compare Gemini-DeepResearch, ChatGPT-DeepResearch, Claude-DeepResearch, and LongCat-DeepResearch on the in-house benchmark. The In-house block of Table 1 reports Content (task coverage and evidential support), Analysis (reasoning and synthesis), and Presentation (organization and clarity), together with the weighted overall score. An automatic evaluator assesses the reports along these dimensions.<sup>1</sup> LongCat-DeepResearch scores 76.04 overall, 0.55 points below ChatGPT-DeepResearch’s 76.59.

## 5 Design Analysis

We study component ablations, ResearchSpec refinement, and Editor scaling on development subsets, and compare model-harness configurations on the full benchmarks. Protocols are reported separately for each study. DRB-I, DRB-II, and RR abbreviate DeepResearchBench, DeepResearchBench II, and ResearchRubrics, respectively.

### 5.1 Component Ablations

Table 2 compares the complete pipeline (Full) with three component interventions for LongCat. Full uses three Planning Writers, the Planning Judge/Critic/Reviser, parallel Researchers, and the Editor. One Writer → Reviser runs these two stages in sequence and removes the Planning Judge and Critic; it therefore changes the planning procedure as well as

<sup>1</sup>We compared the evaluator’s judgments with annotations from four professional annotators; agreement exceeded 95% on the annotated examples.

Writer count. One whole-report Researcher receives Full’s complete ResearchSpec but uses a single research history to produce the draft. No Editor uses the exact draft saved before editing in its originating Full run.

All conditions within each displayed column use identical question IDs. The LongCat DeepResearchBench II comparison uses the common completed questions across all four arms. Selection and the stopped case are documented in [Appendix B.2](#).

| Configuration               | DRB-II       | ResearchRubrics | Avg.         |
|-----------------------------|--------------|-----------------|--------------|
| Full                        | <b>48.68</b> | <b>79.14</b>    | <b>63.91</b> |
| One Writer → Reviser        | 44.65        | 74.46           | 59.56        |
| One whole-report Researcher | 45.64        | 78.49           | 62.07        |
| No Editor                   | 48.60        | 78.00           | 63.30        |

Table 2: LongCat component scores; higher is better. DRB-II denotes DeepResearchBench II, and → denotes stage order. Avg. is the unweighted mean of the two displayed benchmark scores. Bold marks the highest mean per column. Detailed configurations are given in [Appendix B.2](#).

Full’s advantage over simplified planning and a single whole-report Researcher supports the division between compact global coordination and detailed local investigation. ResearchSpec can guide the report’s shared agenda while each section develops its evidence in an independent context.

## 5.2 ResearchSpec Refinement

We examine how planning aggregation and refinement affect final reports on a fixed ResearchRubrics development subset with LongCat. W0 uses a fixed single Writer’s ResearchSpec; J0 integrates three Writers through the Judge, before any Critic/Reviser cycle. R1 and R2 apply one and two successive Critic/Reviser cycles to J0. Each stage runs the same downstream Researcher and Editor configuration; planning compaction is disabled.

[Table 3](#) compares planned coverage with final-report quality; the evaluation protocols are given in [Appendices B.4](#) and [B.6](#). Plan aggregation improves final-report quality, while refinement further improves planned coverage. ResearchSpec thus provides a compact proxy for early report iteration: the system can refine what the report needs to establish before expanding it into full prose.

| Stage | ResearchSpec | ResearchRubrics |              |              |
|-------|--------------|-----------------|--------------|--------------|
|       | Coverage (%) | Explicit        | Implicit     | Total        |
| W0    | 53.76        | 84.34           | 82.21        | 81.67        |
| J0    | 60.47        | 89.09           | 87.11        | <b>85.13</b> |
| R1    | <b>63.14</b> | <b>91.59</b>    | 82.76        | 84.34        |
| R2    | 62.48        | 90.91           | <b>87.99</b> | 84.89        |

Table 3: LongCat planning and report scores on a fixed ResearchRubrics development subset. W0 is the single-Writer plan, J0 the Judge-integrated plan, and R1/R2 one/two Critic/Reviser cycles. ResearchSpec coverage evaluates the plan before report generation ([Appendix B.4](#)). RR Explicit and Implicit score the corresponding report-rubric categories; Total uses the full rubric. Bold marks the best stage for each metric. The protocol is given in [Appendix B.6](#).

## 5.3 Editor Scaling

We study LongCat Editor scaling on fixed development subsets for each benchmark, with comparisons matched by question. A0 is the unedited draft; A1 applies the original Editor once. B1 starts from A0 with an enhanced Editor using source excerpts and fact-checking instructions. B2 and B3 continue the same procedure for two and three rounds, respectively ([Appendix B.7](#)). [Table 4](#) reports native scores and average readability preference for these conditions. The readability evaluation protocol is detailed in [Appendix B.5](#).

Across enhanced Editor rounds, average readability preference improves; the final round also preserves or improves the native benchmark scores relative to the unedited draft. This supports coordinated local revision as a way to develop independently researched sections into a more coherent report. Benchmark-specific preferences and paired results are given in [Appendix B.7](#).

| Configuration                 | Native report score |              |              | Readability     |
|-------------------------------|---------------------|--------------|--------------|-----------------|
|                               | DRB-II              | RR           | Avg.         | Avg. preference |
| A0: No Editor                 | 55.87               | 85.91        | 70.89        | —               |
| A1: Original Editor (1 round) | 56.04               | 85.66        | 70.85        | 50.00           |
| B1: Enhanced (1 round)        | 55.15               | 85.18        | 70.17        | 50.42           |
| B2: Enhanced (2 rounds)       | 55.68               | <b>86.87</b> | 71.28        | 53.33           |
| B3: Enhanced (3 rounds)       | <b>58.38</b>        | 86.00        | <b>72.19</b> | <b>53.96</b>    |

Table 4: LongCat Editor scaling. DRB-II (DeepResearchBench II) and RR (ResearchRubrics) are native report scores on the development subsets; their Avg. is the unweighted mean of the two displayed scores. Readability is the mean automatic preference score across the development subsets against the same A0 draft; it is a transformed preference margin, not a win rate. 50 denotes parity ([Appendix B.5](#)). A dash marks A0 as the ungraded reference. Bold marks the highest mean per column, without implying statistical significance.

## 5.4 Model and Harness Configurations

We compare three configurations on the full benchmark cohorts. The first uses the previous LongCat release with bounded ReAct research followed by a direct report. The second pairs the same previous release with the current harness. The third pairs the current LongCat model with the current harness. ReAct uses the same search and page-reading services and a 65,536-token final-output ceiling, with thinking enabled.

[Table 5](#) reports native report quality for the three configurations and their unweighted averages across the three benchmarks.

| Model                      | Harness               | DRB-I        | DRB-II       | RR           | Avg.         |
|----------------------------|-----------------------|--------------|--------------|--------------|--------------|
| LongCat (previous release) | ReAct / direct report | 47.65        | 33.13        | 60.33        | 47.04        |
| LongCat (previous release) | Current harness       | 53.58        | 48.68        | 71.90        | 58.05        |
| LongCat (current)          | Current harness       | <b>55.25</b> | <b>51.35</b> | <b>79.83</b> | <b>62.14</b> |

Table 5: Report quality across model and harness configurations. Previous release refers to LongCat-2.0. All three configurations are evaluated on the full benchmarks. The last row is LongCat-DeepResearch and reproduces the main results from [Table 1](#). Avg. is the unweighted mean of the three displayed benchmark scores. Bold marks the highest score per column.

The gains from changing the harness with the model fixed show the value of organizing research around explicit requirements and section-level work. Further gains with the current model support treating model capability and research orchestration as complementary parts of the system.

## 5.5 Case Study: From ResearchSpec to Report

We examine a recorded LongCat trajectory for *AI-enhanced portfolio management* (DeepResearchBench II 012). The user requests a review of classical theories, AI/ML applications, multi-criteria decision making, and portfolio optimization and rebalancing, covering methods through April 2024. The case exposes two concrete operations: revising the research agenda before section research, and coordinating overlapping evidence after the section drafts have been assembled. The excerpts below show the plan, section draft, editorial directive, and final report from this trajectory.

### 5.5.1 Turning the Question into Research Assignments

The merged ResearchSpec decomposes the four requested themes into 19 subsections: ten for classical theories, five for AI/ML applications, two for multi-criteria methods, and two for optimization and rebalancing. Each subsection contains *What to Cover*, research questions, required entities or cases, and source leads. These fields separate the topic of a section from the evidence that its Researcher must seek. The specification also records shared boundaries, including the publication cutoff and the priority given to primary, peer-reviewed sources.

### Planning: inspect the agenda before section research

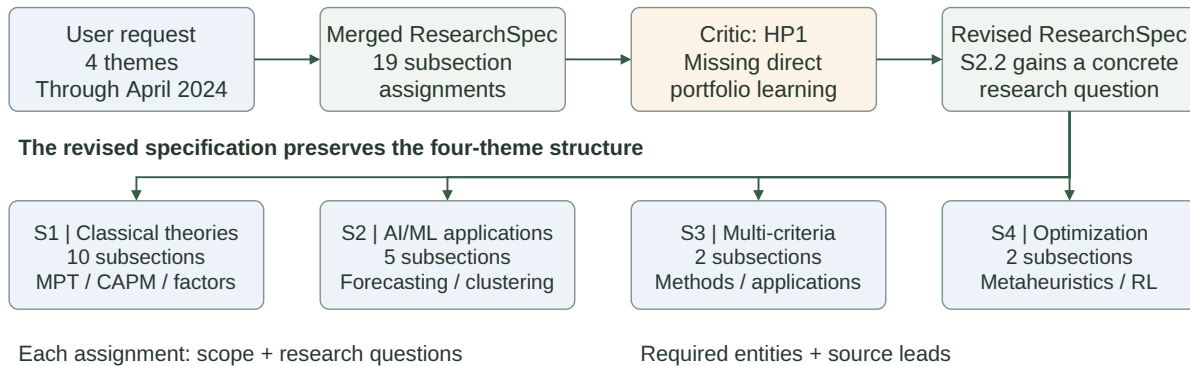


Figure 4: Recorded planning artifacts for the portfolio-management case. The Critic identifies a missing research direction; the Reviser adds it to S2.2 while retaining the 19-subsection structure. The lower row groups the resulting assignments by the user’s four themes. Selected artifacts are shown, not every internal model call.

**A concrete revision before research.** The merged plan covers forecasting methods but does not name Heaton, Polson, and Witte or “Deep Portfolio Theory.” The Critic flags this omission as HP1 and directs attention to learning portfolio weights from data, alongside the existing forecasting agenda. The Reviser adds the topic to S2.2, including a specific research question and a named entity for subsequent investigation. The change expands an existing assignment; it does not create an additional subsection.

| Planning revision                                                                                                                                                                                                                                 | S2.2   Time series forecasting |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>CRITIC: IDENTIFY THE MISSING DIRECTION</b><br>HP1: Heaton, Polson & Witte (2017) — “Deep Portfolio Theory”                                                                                                                                     |                                |
| <b>REVISER: MAKE THE NEW QUESTION EXECUTABLE</b><br>How does Heaton, Polson & Witte (2017) “Deep Portfolio Theory” propose learning portfolio weights directly from data, and how does this differ from the predict-then-optimize approach? (HP1) |                                |

**From an assignment to a section artifact.** The Capital Asset Pricing Model (CAPM) assignment (S1.2) asks which evidence supports or challenges the model and supplies a survey as a source lead. Its recorded Researcher output develops that question into a cited section on empirical tests, Roll’s critique, and Fama–French findings. The final report also contains a dedicated “Deep Portfolio Theory” passage under time series forecasting, preserving the added topic. These are written section artifacts; the next stage coordinates their overlapping material.

#### 5.5.2 Coordinating Evidence Across Section Drafts

The Fama–French findings serve two roles in this report. In the CAPM section they provide an empirical challenge to the model; in the three-factor-model section (S1.5) they motivate an alternative account of returns. Independent section writing can therefore repeat a legitimate piece of evidence. The Global Editor reads the assembled draft and assigns the detailed treatment to S1.5, while directing S1.2 to retain a brief mention and a cross-reference. The Local Editor then applies this decision to the existing CAPM section.

### Preserve the finding, coordinate its detailed treatment



Global reads all section drafts. Only the S1.2 editing path is shown.

Figure 5: The selected CAPM editing path. The Global Editor assigns cross-section ownership; the Local Editor receives that directive and the existing section text. The revised passage retains the empirical finding while referring to S1.5 for its detailed treatment. Other sections in the Global Editor’s assembled context are omitted from this view.

| Observed text change                                                                                                                                                                                                                                                                                                                                                                        | S1.2   CAPM |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>RESEARCH DRAFT</b><br>Using the Fama-MacBeth methodology on NYSE, AMEX, and NASDAQ stocks (1963–1990), they found that the relation between market beta and average return was <b>flat</b> — beta had no explanatory power for the cross-section of average returns when used alone.                                                                                                     |             |
| <b>GLOBAL OWNERSHIP DIRECTIVE</b><br>Fama-French 1992 empirical findings → S1.5; S1.2 cross-references S1.5                                                                                                                                                                                                                                                                                 |             |
| <b>LOCAL REVISION RETAINED IN THE FINAL REPORT</b><br><i>Fama-French (1992).</i> The most damaging empirical challenge to the CAPM came from Fama and French’s (1992) finding that market beta had no explanatory power for the cross-section of average returns, while size and book-to-market equity were strongly and robustly related to average returns (discussed in detail in S1.5). |             |

**Cross-section coordination.** The plan, draft, directive, and final passage can be linked through the same section identifier. The revision operates on an existing research artifact: it preserves the central empirical finding, condenses its local treatment, and records where the detailed discussion belongs. The Global plan makes similar ownership decisions for the “error maximization” concept and for Non-dominated Sorting Genetic Algorithm II (NSGA-II) details, showing that coordination concerns the allocation of substantive material across sections, as well as sentence-level editing. Those directives are recorded decisions; the CAPM passage above is the specific execution traced here.

**Remaining limits.** The example also exposes a finalization mismatch: the report retains the internal “S1.5” reference while the displayed three-factor-model heading omits that identifier. The final report still misses some benchmark requirements, including Treynor and CPPI. Thus, an executable research agenda and a traceable edit do not guarantee complete coverage or flawless presentation. Only the final report was scored; the case does not isolate planning or editing gains. Further excerpts and a contrasting labor-market case with unresolved coverage gaps appear in [Appendix C](#).

## 6 Conclusion

We presented LongCat-DeepResearch, a model-and-harness system that shifts early research iteration from full reports to ResearchSpec. Parallel planning defines evidence requirements and section responsibilities. Researchers develop citation-bearing sections independently, and coordinated Editors revise the assembled draft. These interfaces also support research-task and trajectory construction.

The system achieves 55.25 on DeepResearchBench, 51.35 on DeepResearchBench II, and 79.83 on ResearchRubrics. On matched development subsets, the full LongCat pipeline has the highest component-study means. The full-benchmark model-harness comparison favors the current system: the current harness improves scores with the previous release, and the current model further improves scores under that harness. Further planning refinement produces mixed category-level effects, while enhanced Editor rounds improve average readability preference. These observations are specific to the evaluated configurations and do not isolate training-stage contributions.

ResearchSpec coverage supports early inspection of planned requirements; factual verification and independent report evaluation remain necessary.

## Limitations

**Evaluation and automatic assessment.** Native benchmark scores, planned coverage, and citation support measure different properties. In the automatic benchmark and component evaluations, unchanged content can receive different rubric judgments. Question-bootstrap intervals capture question variation, not uncertainty from repeated generation or judging. Coverage diagnostics support inspection of planned requirements; factual verification and independent report evaluation remain necessary.

**Development evidence and reproducibility.** The component, planning-refinement, and Editor studies use previously inspected development subsets, some selected using prior model scores. Retrieval windows, realized computation, and recovery histories vary across studies. Broader task coverage and repeated runs would help characterize trend stability and resource requirements. Research-task and trajectory construction belong to a broader model development process. The model comparisons evaluate the resulting checkpoints; matched training ablations would help identify contributions from individual data sources and training stages. Changing web content and hosted research products also limit exact replay.

**Planning and editorial scope.** ResearchSpec is refined during planning and fixed during section research. Researchers can investigate further within their assignments, but reopening the global agenda after drafting remains outside the core pipeline. Complete section artifacts support assembly and inspection, yet research and editing can omit relevant details. The framework prioritizes evidence recall and coverage of research requirements. The resulting reports can be lengthy, dense, or repetitive, leaving room to improve readability for human readers. Global and Local Editors read the full draft and remain subject to input-context limits. Additional refinement has varying effects across configurations and benchmarks, consuming resources that could also support evidence gathering. The enhanced Editor jointly changes source access, instructions, and iteration count, leaving their individual contributions unresolved.

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## A Official Leaderboards and Evaluated Systems

Tables 6 and 7 place the official GPT-5.5 (medium) leaderboard snapshots accessed on September 14, 2026 alongside the four systems evaluated in this work. The lower groups reproduce Table 1; they retain their own system snapshots and evaluation settings and are not official leaderboard entries.

| System                                                    | Comp. | Insight | Inst. | Read. | Overall      |
|-----------------------------------------------------------|-------|---------|-------|-------|--------------|
| <i>Official leaderboard snapshot (September 14, 2026)</i> |       |         |       |       |              |
| ATH-Voicepica-DeepResearch                                | 56.37 | 56.77   | 55.92 | 53.27 | 55.99        |
| cellcog-max                                               | 56.34 | 57.08   | 55.30 | 51.94 | 55.78        |
| SPIA DeepResearch                                         | 55.26 | 55.99   | 55.12 | 52.70 | 55.12        |
| WhaleCloud-DocChain_0612                                  | 55.14 | 55.33   | 54.85 | 52.48 | 54.78        |
| deep-dog-2                                                | 54.68 | 55.32   | 54.26 | 51.05 | 54.32        |
| bodhi                                                     | 54.15 | 54.60   | 54.41 | 51.87 | 54.07        |
| lunon_full100_FINAL.submission                            | 53.42 | 54.83   | 53.41 | 50.48 | 53.51        |
| dalpha-deepresearch                                       | 52.58 | 52.94   | 53.87 | 53.20 | 53.10        |
| sourcery                                                  | 50.53 | 52.22   | 51.06 | 49.68 | 51.17        |
| Infosys-MARS-Deepresearch-Agent                           | 50.96 | 51.79   | 51.77 | 48.62 | 51.12        |
| gemini-2.5-pro-deepresearch                               | 50.01 | 49.92   | 50.22 | 49.58 | 49.98        |
| openai-deepresearch                                       | 48.05 | 46.69   | 49.29 | 47.62 | 47.84        |
| perplexity-Research                                       | 41.78 | 41.27   | 45.31 | 46.03 | 43.05        |
| grok-deeper-search                                        | 39.65 | 38.12   | 44.62 | 45.72 | 41.22        |
| <i>Systems evaluated in this work</i>                     |       |         |       |       |              |
| Gemini-DeepResearch                                       | 49.47 | 50.73   | 50.88 | 49.28 | 50.21        |
| ChatGPT-DeepResearch                                      | 55.41 | 55.82   | 54.99 | 51.51 | 54.95        |
| Claude-DeepResearch                                       | 53.71 | 53.87   | 53.59 | 51.35 | 53.43        |
| <b>LongCat-DeepResearch</b>                               | 56.21 | 56.48   | 55.18 | 49.96 | <b>55.25</b> |

Table 6: DeepResearchBench RACE results under GPT-5.5 (medium) evaluation. Comp., Inst. and Read. denote comprehensiveness, instruction following and readability. Official rows preserve their published identities; the lower group uses the system snapshots in Section 4.2.

The Gemini row in our evaluation covers 99 scored tasks; the other three rows cover 100. The newer web-product snapshots and their tool budgets differ from those of the published entries.

| System                                                    | Valid   | Info. | Analysis | Pres. | Overall      |
|-----------------------------------------------------------|---------|-------|----------|-------|--------------|
| <i>Official leaderboard snapshot (September 14, 2026)</i> |         |       |          |       |              |
| GPT-o3 Deep Research                                      | 131/132 | 38.05 | 46.37    | 83.49 | 43.00        |
| Gemini 3 Pro Deep Research                                | 129/132 | 36.58 | 43.91    | 84.94 | 41.40        |
| Gemini 2.5 Pro Deep Research                              | 132/132 | 34.28 | 49.11    | 77.98 | 39.98        |
| Doubao Deep Research                                      | 127/132 | 32.58 | 44.12    | 71.53 | 37.36        |
| Perplexity Research                                       | 127/132 | 29.16 | 36.26    | 73.41 | 33.65        |
| <i>Systems evaluated in this work</i>                     |         |       |          |       |              |
| Gemini-DeepResearch                                       | 131/132 | 41.75 | 51.28    | 84.81 | 46.72        |
| ChatGPT-DeepResearch                                      | 132/132 | 41.97 | 52.26    | 86.98 | 47.16        |
| Claude-DeepResearch                                       | 132/132 | 44.37 | 51.15    | 81.02 | 48.18        |
| <b>LongCat-DeepResearch</b>                               | 131/132 | 46.42 | 60.69    | 83.13 | <b>51.35</b> |

Table 7: DeepResearchBench II results under GPT-5.5 (medium) evaluation. Valid denotes scored tasks; Info. and Pres. denote information recall and presentation. The upper group is the official text-only snapshot; the lower group reproduces our main results.

The combined presentation places our system-level comparison alongside published results without establishing an official leaderboard rank.

**Official snapshot provenance.** The published entries come from [the official leaderboard](#), Hugging Face Space revision ad8fc70b0e6c, using `data_gpt55/leaderboard.csv` and `data_drb2/leaderboard.csv`. The latter is the GPT-5.5 text-only reevaluation of public reports, excluding PDF-only submissions and ignoring embedded DOCX images. The displayed scores retain the dated snapshot values.

## B Experimental Details and Supplementary Results

### B.1 Main Evaluation and Shared Conventions

The LongCat DeepResearchBench result covers 100 scored tasks; ResearchRubrics covers 101 tasks and 2,593 criteria, with thinking enabled, three planning writers, twenty maximum tool rounds and research concurrency three. The selected DeepResearchBench II result uses 131 scored reports out of 132 assigned tasks, excluding missing task 116. Its overall and dimension scores share this denominator.

Web-product coverage also varies: Gemini-DeepResearch has 99 scored DeepResearchBench tasks and 131 DeepResearchBench II tasks, whereas ChatGPT-DeepResearch and Claude-DeepResearch have 100 and 132, respectively. ChatGPT-DeepResearch and Gemini-DeepResearch ResearchRubrics scores use their common 100-task cohort. Claude-DeepResearch’s ResearchRubrics overall is an externally reported aggregate; complete per-task grades are unavailable for verification in this work. These cohorts define the scope of the point estimates in [Table 1](#).

**Web search and page access.** The harness accesses public web information through internally managed search and page-reading services. The `web_search` interface returns result titles, URLs, and snippets from the configured search backend; `web_fetch` retrieves and parses selected pages for further reading. These services provide the retrieval infrastructure used by the research agents, while the harness controls when to search and which pages to read.

**Native scoring and recovery.** DeepResearchBench uses the benchmark’s GPT-5.5 (medium) RACE evaluator. The development DeepResearchBench II studies use GPT-5.5 (medium), 50-rubric chunks and the complete rubric denominator. ResearchRubrics uses Gemini 2.5 Pro, binary verdicts, signed criterion weights and the sum of positive weights as denominator. Recovery retains completed report bytes and grades, reuses exact matching responses, and charges failed attempts to the original task. Successful low-score reports are not selectively regenerated. These measurements include recovery rather than estimating first-attempt reliability.

**In-house evaluation.** The in-house comparison uses an automatic evaluator for report content, analysis, and presentation. [Section 4.5](#) reports its agreement with professional human annotations and the resulting evaluation scores.

### B.2 Component Comparison

The LongCat model is evaluated with thinking enabled. The comparison uses fixed ResearchRubrics and DeepResearchBench II development subsets, restricted to questions completed by all four arms. An incomplete single-Researcher case is excluded from the paired comparison. The interventions are defined in [Section 5.1](#). The single Researcher’s tool allowance sums the section allowances within the task ceiling, and memory compression is disabled in the component variants.

The studies have one completed grade per report and do not estimate repeated-Judge variance. Official reference-result alignment remains unestablished for these previously inspected development subsets.

Realized computation and retrieval windows differ across these component conditions. The reported outcomes are native benchmark scores; dedicated fact-support and readability evaluations are not complete for these cohorts.

### B.3 Model-Harness Configuration Settings

All three configurations in [Table 5](#) use the full benchmarks. ReAct/direct reporting has a 65,536-token final-response ceiling, while the current harness in this comparison has a 32,000-token per-call ceiling. Cumulative task limits are 600 model calls, 1,048,576 completion tokens, 500 searches and 500 page reads. These allowances do not imply equal realized compute. Same-model recovery retains validated plans and sections, original spending and unchanged scoring rules. Comparisons across retrieval windows and recovered runs characterize the recorded configurations.

## B.4 ResearchSpec Coverage Evaluation

We evaluate a ResearchSpec before report generation using GPT-5.5 (medium). The evaluator compares the plan with the same question’s rubric and maps each criterion to supporting plan excerpts and section identifiers; Researcher and Editor are not executed for this diagnostic. For each question, coverage is the percentage of positive-weight criteria judged fully covered, with criteria counted equally. Partial, missing, and not-plan-evaluable criteria remain in the denominator. We report the mean of these per-question percentages over the evaluation subset. This score measures coverage of the full positive-criterion set by the plan, rather than final-report quality or factual correctness. Including criteria that cannot be evaluated from a plan limits its interpretation as a measure of planning completeness; comparisons use the same criterion set. Rubrics and diagnostic judgments are withheld from the generating agents.

## B.5 Readability Preference Evaluation

GPT-5.5 (medium), with thinking enabled and seed 42, compares each edited report with its own unedited draft (A0) for organization and expression. The evaluator reads both full reports under three evaluation guides and both presentation orders. We orient the margins from both orders toward the edited report and average them for each question using the evaluator’s guide weighting. The resulting margin  $m \in [-2, 2]$  is mapped to  $50 + 25m$ : 50 denotes parity, and higher values favor the edited report. We average the question-level scores within each benchmark, then weight the two benchmark means equally. Every edited condition is compared directly with A0; adjacent-round preferences are not accumulated. A0 is a reference without a self-judgment. These scores express automatic relative preferences, rather than absolute readability or human ratings.

## B.6 ResearchRubrics Refinement Protocol

Table 3 uses the same LongCat development subset across all four stages. For each question, W0 reuses the first Writer from a single three-Writer planning execution; it is not selected by report score. J0 is the Judge’s integrated plan before critique or revision. R1 revises J0 once, and R2 revises R1 once more. The experiment disables planning compaction and explicitly executes both Critic/Reviser cycles. Each of the four immutable planning snapshots is passed to the original parallel Researchers and Editor. Generation uses thinking and seed 42 with the original cumulative task budgets. Comparisons use matched question IDs, and results are averaged over questions.

ResearchSpec coverage follows Appendix B.4; the comparison reuses the existing coverage judgments.

**Explicit and implicit requirements.** We group the original ResearchRubrics criteria by their supplied *Explicit Criteria* and *Implicit Criteria* labels. For each question and category, the score is 100 times the sum of satisfied criteria’s signed weights divided by the sum of positive weights in that category. The table reports the mean of these per-question scores; both categories are present throughout the development subset. Negative-weight criteria retain their penalties. Total uses all rubric categories and their full positive-weight denominator, so it is not the average of the two displayed subscores. Category scores are computed from the same Gemini 2.5 Pro criterion-level verdicts as the total score. Rubrics and diagnostic judgments are never supplied to the generating agents.

## B.7 Editor Refinement: Settings and Paired Results

LongCat uses fixed development subsets for each benchmark and five conditions. Thinking is enabled and seed 42 is requested. The development subsets were expanded after inspection of the initial results; this is exploratory rather than independent confirmation.

A0 is the unedited draft. A1 applies the original Editor once; B1 applies the enhanced Editor once, independently from the same A0. B2 and B3 apply the second and third enhanced rounds, each continuing its predecessor. The enhanced Global Editor receives citation-bound excerpts from already retrieved sources, limited to 24,000 characters in total and 2,400 per source, without new retrieval. Source access, fact-checking instructions and iteration count form a joint intervention. Per-call output permits 64,000 tokens. Cumulative task limits are 600 model calls, 1,048,576 completion tokens, 500 searches, and 500 page reads. Failed stages retain original costs and accepted artifacts. LongCat recovery includes directive-bound URL correction and syntax repair in auxiliary quotation fields, preserving native verdicts and scores.

The native scoring rules follow the shared conventions above. Each condition has one completed grade; official reference-result alignment remains unestablished.

**Readability results.** Readability preference follows [Appendix B.5](#). The aggregate trend need not hold within each benchmark. For example, LongCat B3 has mean margins of  $-0.183$  on DRB-II and  $+0.500$  on ResearchRubrics relative to A0, yielding  $+0.158$  overall.

**Paired results.** [Table 8](#) reports the paired contrasts supporting [Table 4](#). Intervals use 10,000 question-bootstrap draws with seed 42 and are unadjusted for multiple comparisons. LongCat B3–A0 changes DRB-II by  $+2.51$  and ResearchRubrics by  $+0.09$ . These intervals cover question variation, not repeated-generation or repeated-Judge variance.

| Benchmark       | Difference | Mean change | Descriptive 95% interval |
|-----------------|------------|-------------|--------------------------|
| DRB-II          | A1 – A0    | $+0.17$     | $[-0.71, +1.11]$         |
| DRB-II          | B3 – A0    | $+2.51$     | $[+0.63, +4.78]$         |
| DRB-II          | B1 – A1    | $-0.89$     | $[-2.51, +0.56]$         |
| DRB-II          | B2 – B1    | $+0.52$     | $[-0.57, +1.70]$         |
| DRB-II          | B3 – B2    | $+2.70$     | $[+0.51, +5.24]$         |
| ResearchRubrics | A1 – A0    | $-0.25$     | $[-2.08, +1.89]$         |
| ResearchRubrics | B3 – A0    | $+0.09$     | $[-0.79, +1.18]$         |
| ResearchRubrics | B1 – A1    | $-0.48$     | $[-3.00, +1.89]$         |
| ResearchRubrics | B2 – B1    | $+1.69$     | $[-0.03, +3.59]$         |
| ResearchRubrics | B3 – B2    | $-0.87$     | $[-2.35, +0.58]$         |

Table 8: LongCat paired native-score changes on fixed development subsets. A0 is the unedited draft, A1 one original Editor pass, and B1–B3 one to three enhanced Editor rounds. Positive changes favor the first condition. DRB-II denotes DeepResearchBench II.

**What the edits change.** On DRB-II 58, editing corrects the online-learning study table from “Arab Region” and 3,348 participants to “Jordan” and “280 students + 50 faculty,” consistent with the detailed discussion. The native score rises by 1.59 points. On the universal basic income task, clearer pilot-specific sections gain a structural criterion, but weaker attribution of quantitative claims loses a higher-weight citation criterion, yielding  $-1.25$  points. These examples show a factual correction in one report and a loss of attribution quality in another.

Some score changes reflect Judge sensitivity: most newly credited hydroforming facts were already in the unedited draft (A0). A higher score therefore need not indicate that editing added the credited information. No model parameters are updated in this study.

## C Case Studies: From ResearchSpec to Edited Report

English-language examples from the selected DeepResearchBench II cohort. Evaluated with GPT-5.5 (medium).

In the excerpts,  $Sx.y$  identifies subsection  $y$  of section  $x$ ; HP1 labels the Critic’s first listed issue. MPT denotes modern portfolio theory, CPPI constant-proportion portfolio insurance, and CI confidence interval.

### CASE 01 AI-enhanced portfolio management

DeepResearchBench II 012

FINAL SCORE **89.29 / 100**

#### TASK & SCOPE

Review classical theories, AI/ML applications, multi-criteria decision making, and portfolio optimization and rebalancing through April 2024.

Task synopsis. The complete ResearchSpec assigns 19 subsections.

#### RESEARCHSPEC

research\_spec.md / ReportSpec / selected original passages

#### Global research boundaries

Output language: English (matching the research question).

[...]

#### S1.2 | Capital Asset Pricing Model (CAPM)

Within S1: Classic Theoretical Frameworks

## CASE 01 AI-enhanced portfolio management (*continued*)

### What to Cover

Present CAPM as the equilibrium asset pricing model building on MPT. Explain the core idea: expected excess return of any asset is proportional to its beta (systematic risk) relative to the market portfolio.

[...]

### Research Questions

Who originated CAPM and in what years (Sharpe 1964, Lintner 1965, Mossin 1966)?

[...]

### Required Entities / Cases

[...]

Jan Mossin (1966)

Security Market Line (SML)

[...]

### Source Leads

<https://www.investopedia.com/terms/c/capm.asp> - CAPM overview (orientation only; find primary peer-reviewed source)

[...]

## FINAL EVALUATION

The plan still omits Treynor (1962); the final report also misses CPPI and fails the chronological-order criterion.

Source text is abridged, not rewritten. Field labels and Markdown are typeset for readability; [...] marks omitted passages. The panels show selected ResearchSpec fields and source passages.

## CASE 01 From the plan to the edited report

Recorded stage changes

DeepResearchBench II 012

### 01 PLANNING REFINEMENT

#### Critic identifies a missing method

HP1: Heaton, Polson & Witte (2017) — "Deep Portfolio Theory"

#### Reviser adds a research question in S2.2

How does Heaton, Polson & Witte (2017) "Deep Portfolio Theory" propose learning portfolio weights directly from data, and how does this differ from the predict-then-optimize approach? (*HP1*)

Heaton et al. is absent from the merged spec and present in the revised spec. This changes the research agenda before section writing.

### 02 RESEARCH DRAFT

S1.2 / CAPM researcher / excerpt before editing

Using the Fama-MacBeth methodology on NYSE, AMEX, and NASDAQ stocks (1963–1990), they found that the relation between market beta and average return was **flat** — beta had no explanatory power for the cross-section of average returns when used alone.

### 03 GLOBAL EDIT

Ownership decision after the section drafts are assembled

Fama-French 1992 empirical findings → S1.5; S1.2 cross-references S1.5

### 04 LOCAL EDIT → FINAL REPORT

S1.2 / revised passage

*Fama-French (1992)*. The most damaging empirical challenge to the CAPM came from Fama and French's (1992) finding that market beta had no explanatory power for the cross-section of average returns, while size and book-to-market equity were strongly and robustly related to average returns (discussed in detail in S1.5).

This passage survives in the scored final report. The CAPM discussion keeps the principal finding and points to S1.5 for the detailed treatment.

## CASE 01 From the plan to the edited report (*continued*)

### WHAT CHANGED

The plan gains a specific research question. Section research expands assignments into cited prose. Editing coordinates repeated material across sections through an ownership decision and a shorter cross-reference.

### What this does not establish

Only the final report has a benchmark score. This trace does not measure the score contribution of planning or editing.

Quotes retain the recorded wording. Commentary describes observed text changes, not measured intermediate score gains.

## CASE 02 Generative AI and the labor market

DeepResearchBench II 056

FINAL SCORE **11.43 / 100**

### TASK AND RESEARCH AGENDA

Review pre-June 2023 literature on positive views, negative views, challenges and opportunities, followed by a study-level application table. The ResearchSpec contains fourteen subsections across these themes and a separate summary-table section.

### CRITIC → REVISED SPEC

HP1: Peng et al. (2023) — GitHub Copilot Developer Productivity Study

The merged Spec does not name Peng et al. or GitHub Copilot. The revised Spec adds the study, its arXiv source lead, and a research question:

What productivity effect did Peng et al. (2023) document for software developers using GitHub Copilot, and what does the heterogeneous-effects finding suggest about career transitions into software development? [HP1]

### RESEARCH → FINAL REPORT

The treatment group completed the task **55.8% faster** than the control group (71.17 minutes vs. 160.89 minutes on average,  $p = 0.0017$ , 95% CI: 21–89%).

This sentence appears in the Researcher output and survives editing in the scored final report. Global editing also assigns the aggregate labor-exposure figures to S2.1, and the Local Editor applies that cross-reference.

### REMAINING COVERAGE GAP

The 121,366-character final report emphasizes aggregate labor-market exposure and prominent productivity studies. It omits the rubric-required studies *Can AI help for scientific writing?* and *Comparing physician and AI chatbot responses to patient questions posted to a public social media forum*. Both titles are absent from the revised Spec and final report, and information recall remains zero.

### TAKEAWAY

The planning addition and its downstream passage are traceable, while the remaining coverage gap persists. This case illustrates how refining an existing agenda can leave other required evidence uninvestigated; its final score does not isolate the contribution of an individual stage.

Quotes are abridged from the planning, research, and editing outputs.

## D Full Author List

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